

Solution Architecture

Date	06 July 2026 – 07 July 2026
Team ID	RW-2026-01
Project Name	Rising Waters
Maximum Marks	5 Marks

Component Description Table

Component Name	Description / Role in Architecture	Technologies Used
Presentation Layer	Provides a responsive web interface for users and administrators to enter weather parameters, view flood predictions, and receive alerts.	HTML5, CSS3, Bootstrap, JavaScript
API Gateway / Application Server	Receives user requests, routes them to the machine learning services, processes prediction requests, and returns the results.	Python, Flask
Authentication Service	Manages user login, session handling, and secure access to the application.	Flask Sessions, Python
Core Logic Service	Performs data preprocessing, feature scaling, machine learning prediction using XGBoost, and flood risk assessment.	Python, Pandas, NumPy, Scikit-learn, XGBoost
External API Integration	Retrieves weather forecasts, rainfall information, and river water level data from external sources for real-time prediction.	Weather APIs, Government Data APIs
Data / Storage Layer	Stores historical datasets, trained machine learning models, prediction results, and alert logs required by the application.	CSV Files, Joblib (floods.save, transform.save)

Solution Architecture Diagram:

The Rising Waters solution architecture follows a layered design consisting of the Presentation Layer, Application Layer, and Data Layer. The web interface enables users to access flood predictions and alerts, while the backend processes weather data using machine learning models. The system integrates external weather data sources and securely stores historical data and prediction models to provide accurate and real-time flood forecasting.

